

onero Robotic Gripper User Guide

Cautions

- Use the gripper only within the specified operating voltage and working environment. Operation beyond specifications may damage the product.
- Before use, check that the gripper body, fingertips, cables, communication terminals, and mounting structure are intact. If any looseness, damage, aging, or abnormal condition is found, stop using the product and contact maintenance personnel.
- Do not place fingers or other body parts within the gripper opening and closing range to avoid pinch injuries.
- Gripping performance is affected by stroke, object material, surface friction, shape, and center of gravity. Use the gripper within the recommended load range whenever possible.
- The gripper uses a lead-screw transmission design. If the motor is powered off or disabled, the held object may fall. Do not rely on power-off holding capability in high-risk scenarios.
- Do not disassemble the gripper, fingertip tactile sensors, or internal transmission mechanism without authorization, as this may affect gripper accuracy, tactile measurement accuracy, and calibration results.

Product Features

- Two-finger parallel gripper structure, suitable for robotic grasping, handling, manipulation, and data collection scenarios.
- Total weight below 500 g, suitable for lightweight robotic arm end-effector deployment.
- Maximum recommended payload of 3.0 kg, covering common desktop object grasping requirements.
- Maximum gripping force of 40 N; higher gripping force can be achieved at certain stroke positions to improve grasp stability.
- Opening/closing time ≤ 1.0 s, meeting fast pick-and-place requirements.
- Repeatability of ± 1.0 mm for stable gripper position control.
- Supports position mode, torque mode, and MIT mode for different control strategies.
- Supports zero-point calibration, fault protection, and tactile sensor data reporting.
- Supports CAN communication with an XT30(2+2) communication terminal.
- Supports OTA upgrade via CAN.

- Optional fingertip tactile sensors are available for contact force perception and multimodal manipulation data collection.

Product Overview

The onero Robotics Gripper is a two-finger parallel gripper module for robotic platforms. It combines lightweight design, grasping capability, positioning accuracy, and communication/control capabilities, making it suitable for robotic end-effector deployment and manipulation data collection. Typical use cases include:

- Robotic end-effector grasping and handling.
- Desktop object picking, moving, and placing.
- Position control, torque control, and constant-force control.
- Gripper width, tactile, IMU, and video data collection.
- Single-arm or dual-arm robotic manipulation data collection.
- Daily manipulation tasks such as folding clothes, opening washing machine doors, and opening dishwasher doors.



Product illustration

Product Specifications

Key Specifications

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Category	Item	Specification
Mechanical	Gripper type	Two-finger parallel gripper
Mechanical	Product weight	500 g
Mechanical	Effective opening width	Approx. 80 mm
Performance	Maximum gripping force	40 N
Performance	Recommended payload	≤ 3.0 kg
Performance	Opening/closing time	≤ 1.0 s
Accuracy	Repeatability	±1.0 mm
Electrical	Rated operating voltage	24 ~ 48 V DC
Communication	Communication protocol	CAN, CANFD
Communication	Communication interface	XT30(2+2)
Interaction	Status indication	Dual-color LED indicator
Upgrade	OTA	Supports OTA upgrade via CAN
Version configuration	Tactile sensor	Available in both non-tactile and tactile-sensor versions

Gripping Force and Payload Capability

Item	Specification / Description
Maximum gripping force	40 N
Recommended payload	≤ 3.0 kg
Gripping-force characteristics	Gripping force varies with opening/closing stroke, control mode, and object contact state.
Typical gripping-force performance	At certain small-to-medium stroke positions, measured gripping force can exceed 80 N.
Grip holding capability	For objects that can be grasped stably, the gripper can hold them for more than 15 min without slipping.

Opening and Closing Speed

Item	Specification
Opening-to-closing time	≤ 1.0 s
Closing-to-opening time	≤ 1.0 s
Typical performance	Opening to closing: approx. 0.92 ~ 0.94 s; closing to opening: approx. 0.93 ~ 0.95 s.

Repeatability

Item	Specification / Typical Performance
Repeatability	± 1.0 mm
Typical positioning deviation	Typical deviation is less than 0.3 mm at different opening positions.

Control and Software Capabilities

Category	Item	Description
Control mode	Position mode	Used for gripper opening, closing, and specified-position control.
Control mode	Torque mode	Used for force-control scenarios such as object gripping.
Control mode	MIT mode	Used for more flexible hybrid force-position control.
Control capability	Constant-force control	Supports constant-force gripping control.
Calibration	Zero-point calibration	Supports gripper zero-point calibration.
Data capability	Tactile sensor data reporting	The tactile version supports tactile sensor data output.
Protection	Fault protection	Supports basic fault protection.
Upgrade	OTA	Supports OTA upgrade via CAN.

Motor and Torque Reference

Item	Typical Data / Description
Calibration voltage	Approx. 24 V
Forward Q-axis current - torque	$1\text{ A} \approx 1\text{ N}\cdot\text{m}$; $3\text{ A} \approx 2.8\text{ N}\cdot\text{m}$; $4\text{ A} \approx 3.6\text{ N}\cdot\text{m}$; $9\text{ A} \approx 7.3\text{ N}\cdot\text{m}$
Reverse Q-axis current - torque	$-1\text{ A} \approx -0.5\text{ N}\cdot\text{m}$; $-3\text{ A} \approx -2.5\text{ N}\cdot\text{m}$; $-4\text{ A} \approx -3.4\text{ N}\cdot\text{m}$; $-9\text{ A} \approx -7.4\text{ N}\cdot\text{m}$
Temperature-rise characteristics	Continuous high-torque holding increases temperature rise; system-level protection should be designed according to the use case.

Fingertip Tactile Specifications (Tactile Version)

Category	Item	Specification
Sensor	Tactile sensing location	Left and right gripper fingertips
Sensor	Number of tactile arrays	9×2 , distributed across the fingertips
Sensor	Minimum detectable force	0.3 N
Range	Normal-force measurement range	0 ~ 25 N
Range	Tangential-force measurement range	$\pm 10\text{ N}$
Output	Tactile data output frequency	100 Hz
Reliability	Safe overload	200% FS
Typical measurement deviation	Less than 0.7 N	

Interface and Communication

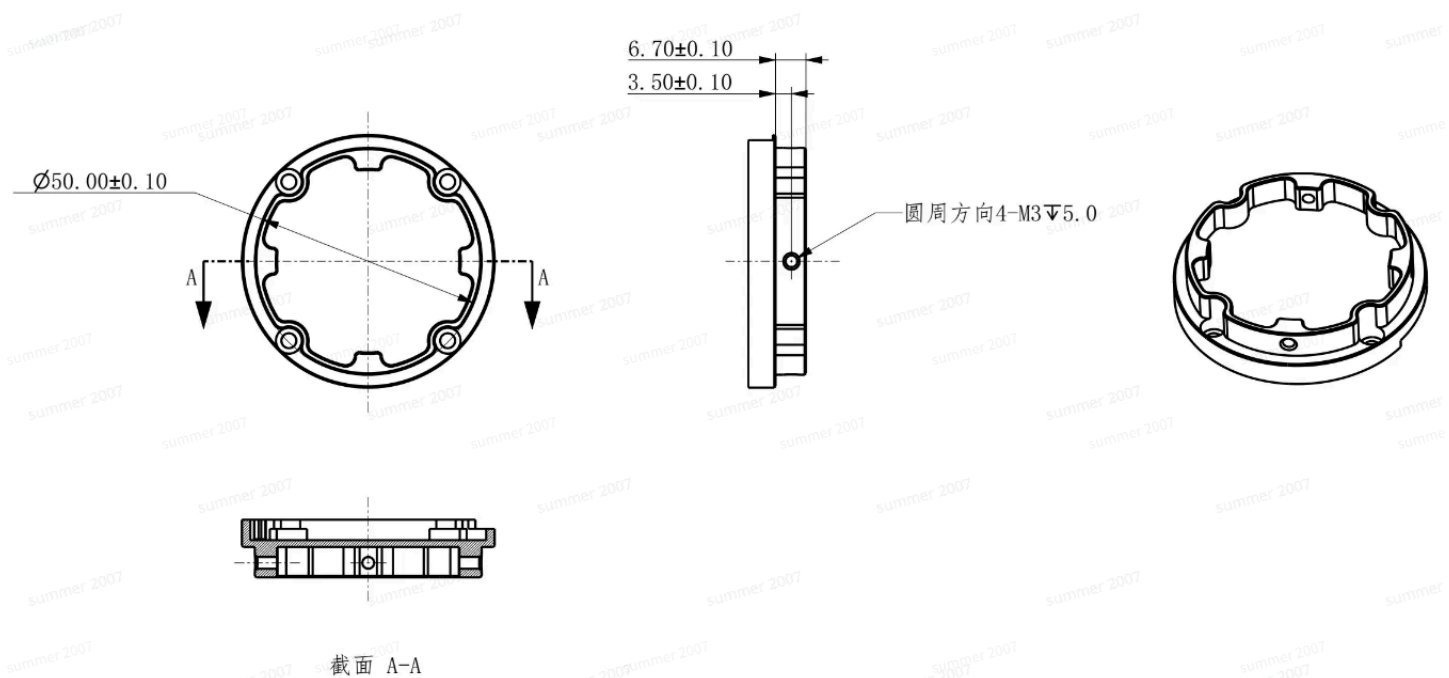
Item	Specification / Description
Communication protocol	CAN

Communication interface	XT30(2+2)
OTA	Supports OTA upgrade via CAN
Control modes	Position mode, torque mode, MIT mode
Sensor communication	The tactile version supports tactile sensor data reporting.

Structure and Installation Notes

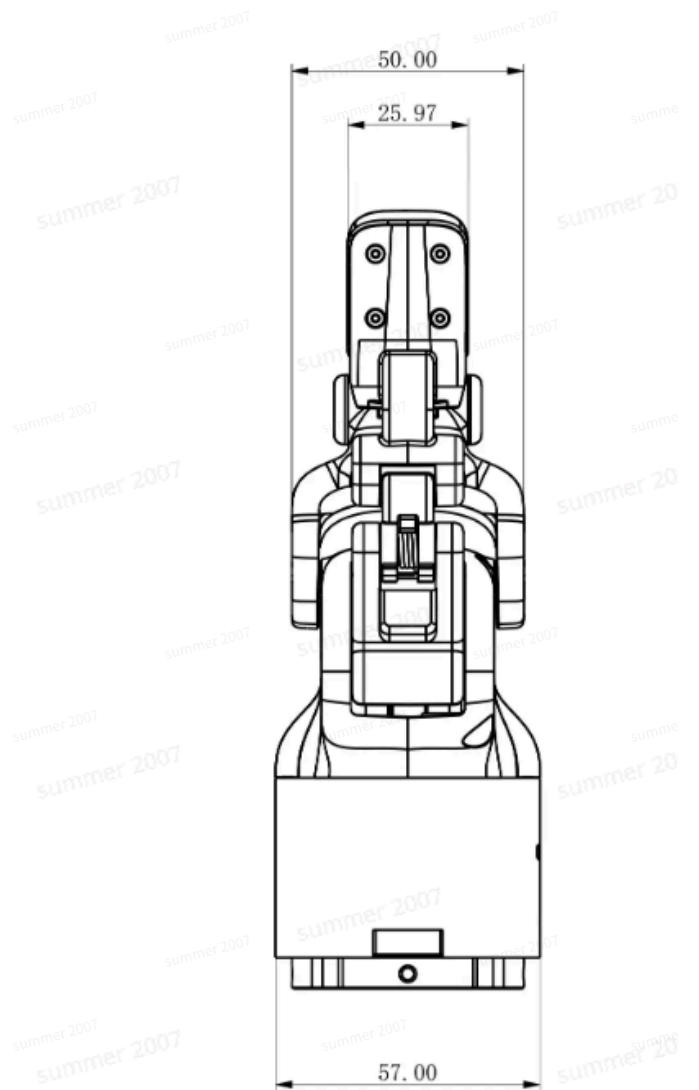
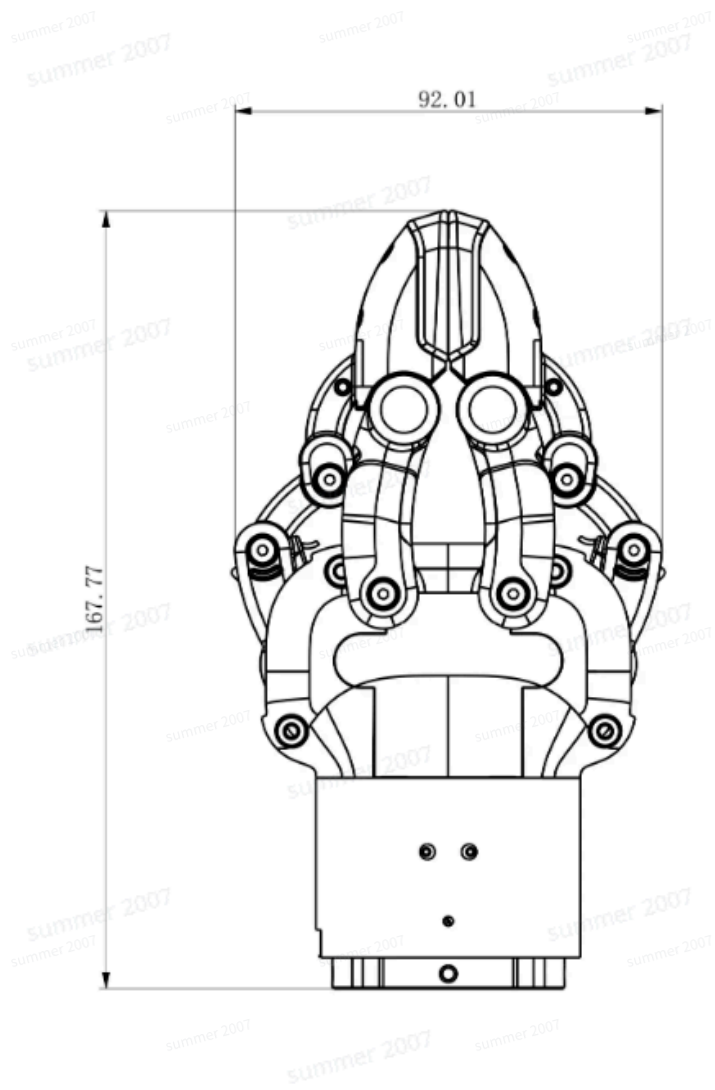
- The gripper uses a two-finger parallel opening and closing structure.
- During installation, ensure the gripper is securely fixed and that cables have sufficient bending allowance to avoid pulling the communication or power cables during motion.
- For high-load, high-frequency, or high-speed motion scenarios, verify the overall robot structural strength, end-effector payload capacity, and motion trajectory before use.

Installation Adapter Dimensions



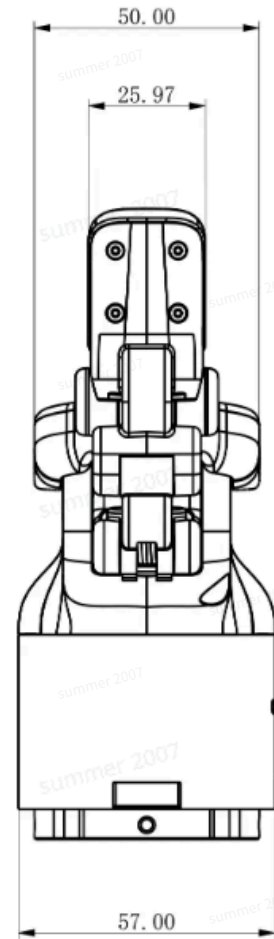
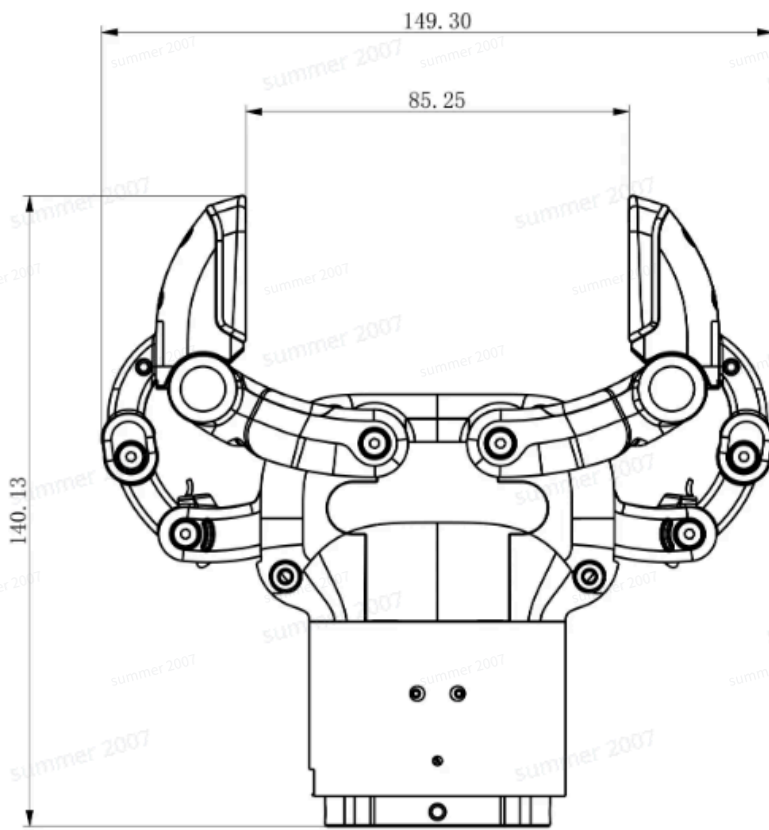
Installation adapter dimensions

Gripper Closed-State Dimensions



Gripper closed-state dimensions

Gripper Open-State Dimensions



Gripper open-state dimensions